Wage Inequality Wars

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UBC

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Today

- Today we will review several papers that have used a "neo-classical" paradigm to explain the rise of wage inequality in the United States that started approximately in 1980.
- We will start by analyzing how an (exogenous?) increase in the supply of a particular group of workers influence the wages of other workers.
- We will then move into the so called SBTC literature. This literature argues that increases in wage inequality during the 1980 is driven by increases in technology that have "pushed up the price" of high skilled workers.
- The next lecture will examine papers that have criticized the SBTC change and analyzed other factors that might have played a role to explain the rise in inequality.

Acemoglu, Autor and Lyle (2004)

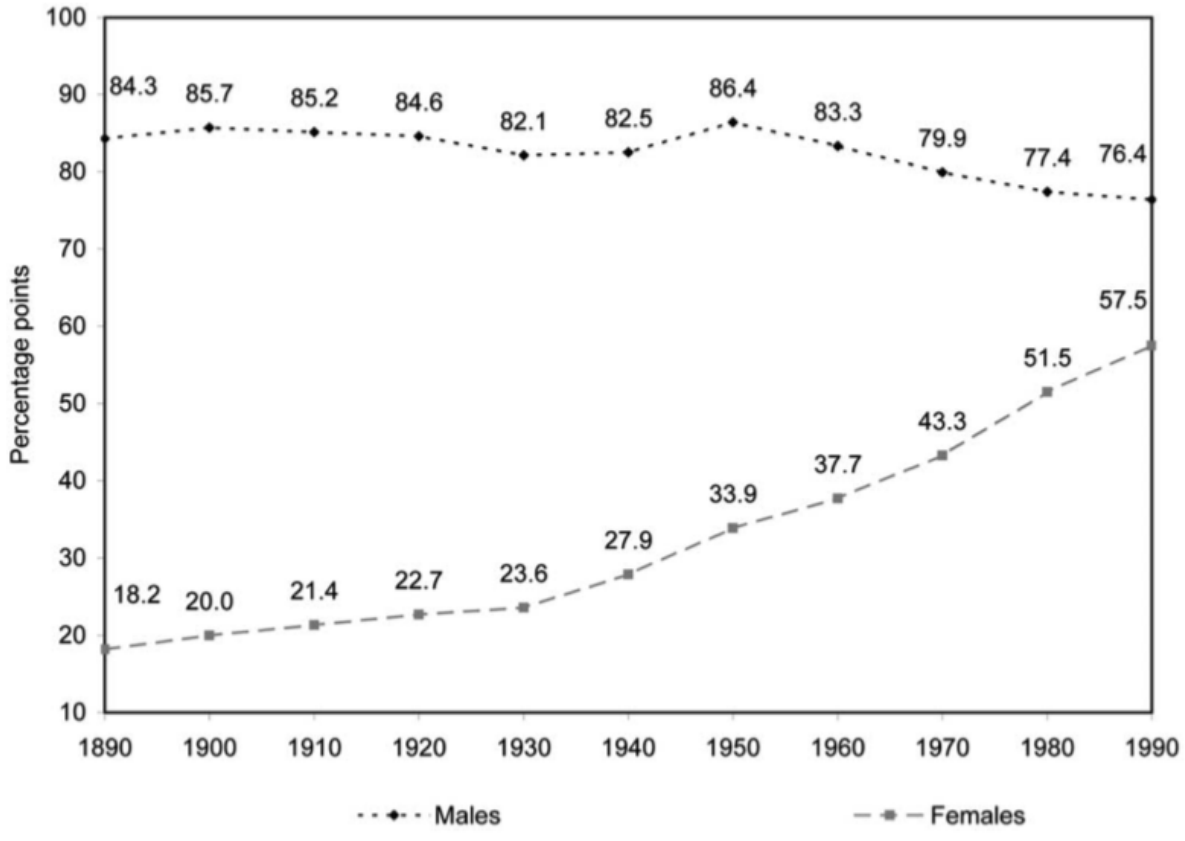
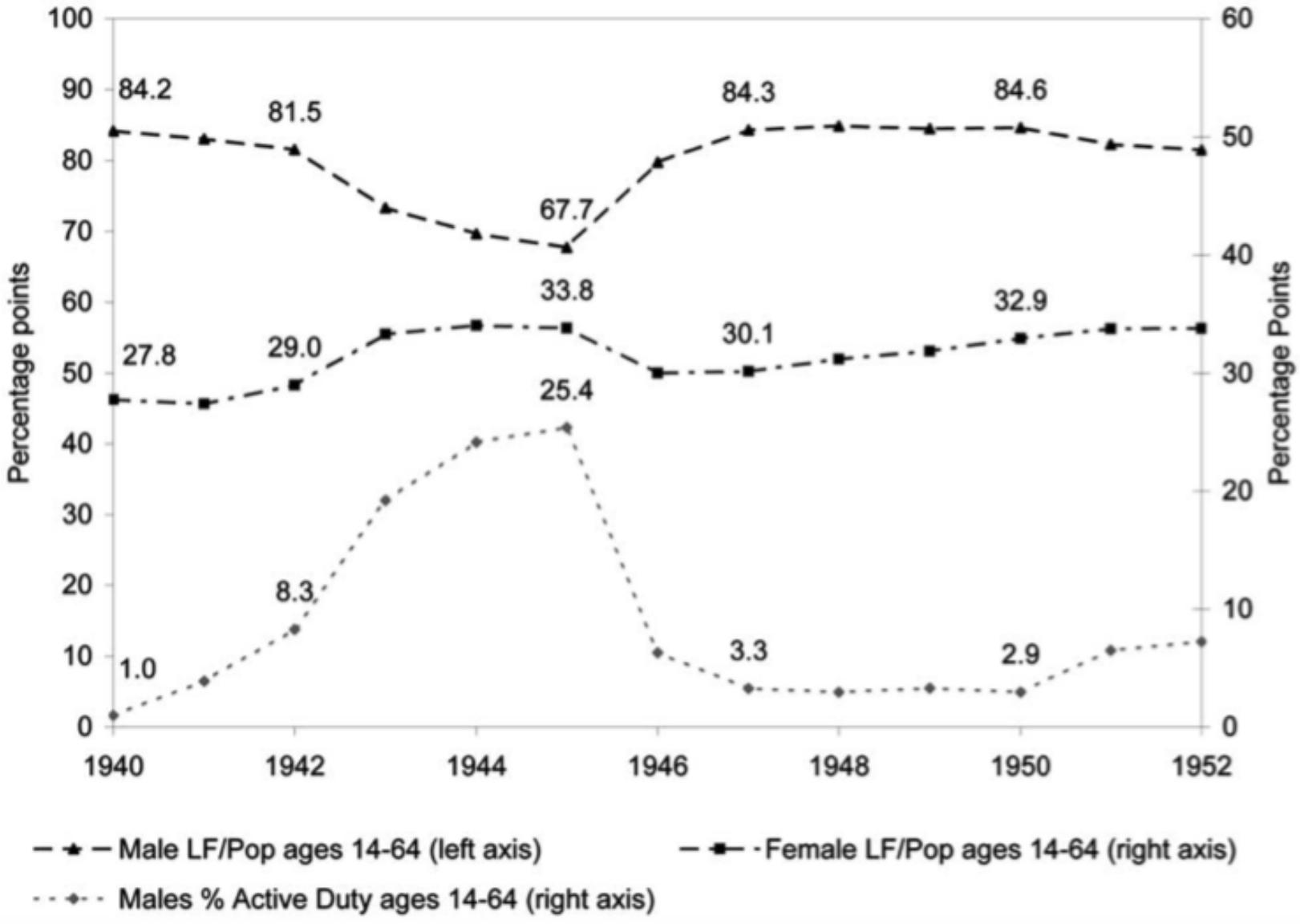


FIG. 1.—Labor force participation by gender of U.S. residents, 1890–1990. Source: Blau, Ferber, and Winkler (2002, table 4.1). Participation rates pertain to the total population prior to 1950 and the civilian population thereafter. Data include individuals 14+ years of age prior to 1950 and 16+ years thereafter.

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- WWII pushed a lot of women into the labor force
- What this (plausibly exogenous) supply shock implied for the earnings of men (and other women)?
- Exploit state level variation in WWII effects on female LFP.
  - Massachusetts: 55 percent of men between the ages of 18 and 44 left the labor market to serve in the war.
  - Georgia: $\approx$ 40-45 percent.

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  - Massachusetts: 55 percent of men between the ages of 18 and 44 left the labor market to serve in the war.
  - Georgia: $\approx$ 40-45 percent.
- Why these differences?
  - Exemptions for farmers; Age composition; Ethic and occupational structures.
  - Country of origin composition (Italy, German, Japanese).
  - Idiosyncratic differences in the behavior of local draft boards.

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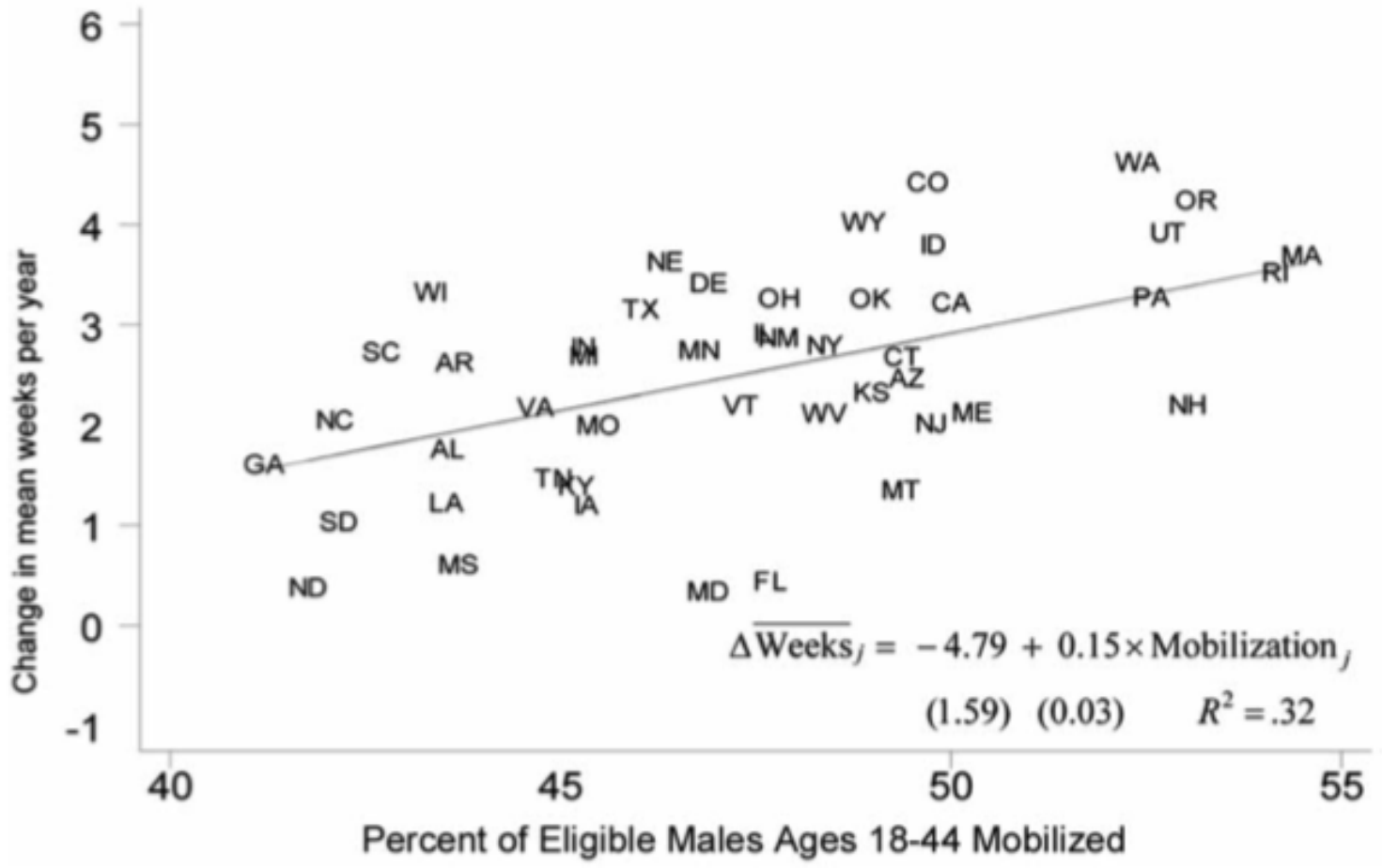


FIG. 4.—State WWII mobilization rates and change in mean female weeks worked per year, 1940–50.

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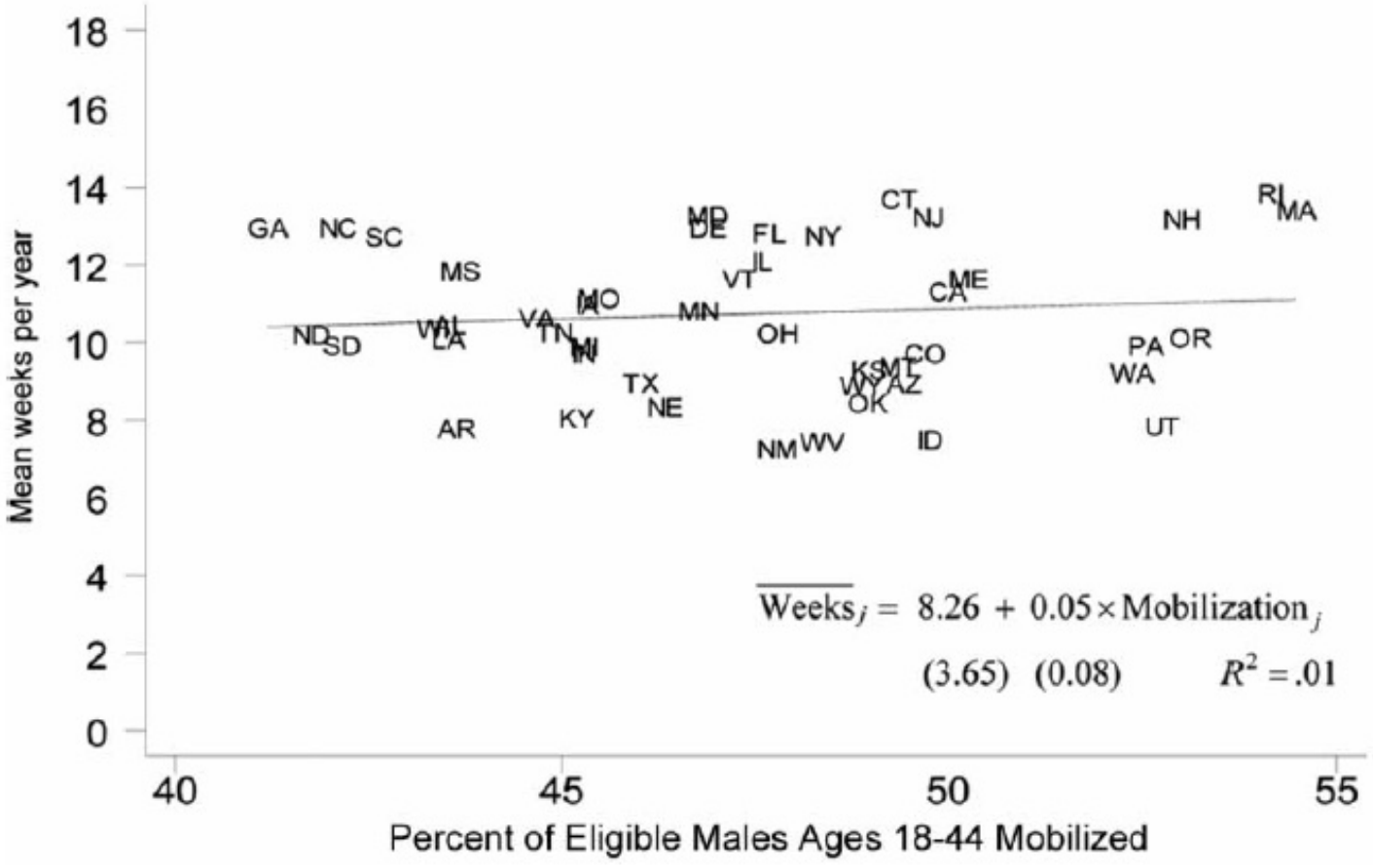


FIG. 3.—State WWII mobilization rates and mean female weeks worked per year, 1940

Model

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- Three factor nested CES

$$Y_t = A_t K_t^{\alpha} \left[(1-\lambda)(B_t^M M_t)^{\rho} + \lambda (B_t^F F_t)^{\rho} \right]^{\frac{1-\alpha}{\rho}} \tag{1}$$

  - $\rho \leq 1$
  - $A_t \equiv$ neutral productivity.
  - B_t^G: biased technical shifter for gender $G \in \{M; F\}$.
  - Elasticity of substitution between L and K?

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  - Elasticity of substitution between L and K?
  - Elasticity of substitution b/w M and F: $\sigma_{MF} \equiv \dfrac{1}{1-\rho}$

- Competitive paradigm implies that

$$w_t^F = (1-\alpha)\lambda B_t^F A_t K_t^{\alpha} (B_t^F F_t)^{-\alpha} \left[(1-\lambda) \left(\frac{B_t^M M_t}{B_t^F F_t} \right)^{\rho} + \lambda \right]^{\frac{(1-\alpha-\rho)}{\rho}} \tag{2}$$

$$w_t^M = (1-\alpha)(1-\lambda) B_t^M A_t K_t^{\alpha} (B_t^M M_t)^{-\alpha} \left[(1-\lambda) + \lambda \left(\frac{B_t^F F_t}{B_t^M M_t} \right)^{\rho} \right]^{\frac{(1-\alpha-\rho)}{\rho}} \tag{3}$$

Derivations

- Want: "short run general equilibrium elasticities" (fixed capital and supply of men)

$$\frac{\partial \log w_t^F}{\partial \log F_t} \equiv \frac{1}{\sigma_F} = -(1 - s_t^m)\alpha - s_t^m \frac{1}{\sigma_{MF}} \tag{4}$$

where $s_t^m = \dfrac{w_t^M M_t}{w_t^M M_t + w_t^F F_t}$.

$$\frac{\partial \log w_t^M}{\partial \log F_t} \equiv \frac{1}{\sigma_M} = -(1 - s_t^m)\alpha + (1 - s_t^m)\frac{1}{\sigma_{MF}} \tag{5}$$

- Case 1: $\sigma_{MF} = \infty \rightarrow$ male and female wages both have elasticity $-(1 - s_t^m)\alpha$. Intuition?
- Case 2: $\alpha = 0$ additional females raises men wages. Intuition?

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- Case 2: $\alpha = 0$ additional females raises men wages. Intuition?
- Anything weird about these calculations?

Adding quality

- Expand the production function as follows now

$$Y_t = A_t K_t^{\alpha} \left\{ (B_t^L)^{\zeta} + \underbrace{\left[(B_t^F F_t)^{\mu} + (B_t^H H_t)^{\mu} \right]^{\frac{\zeta}{\mu}}}_{N_t} \right\}^{\frac{1-\alpha}{\zeta}} \tag{6}$$

  - $\sigma_{FH} = (1-\mu)^{-1}$; $\sigma_{FN} = (1-\zeta)^{-1}$.
  - When $\zeta > \mu$ female labor competes more with low skill male labor than high skill male labor.

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  - $\sigma_{FH} = (1-\mu)^{-1}$; $\sigma_{FN} = (1-\zeta)^{-1}$.
  - When $\zeta > \mu$ female labor competes more with low skill male labor than high skill male labor.
- We have now the technology to understand the implications of $\uparrow F_t$ on the male skill premium:

$$\omega_t = \frac{w_t^H}{w_t^L} = \frac{B_t^H}{B_t^L} \frac{\left(B_t^H H_t\right)^{\mu-1} \left[(B_t^F F_t)^{\mu} (B_t^H H_t)^{\mu} \right]^{\frac{\zeta-\mu}{\mu}}}{(B_t^L L_t)^{\zeta-1}} \tag{7}$$

- This implies that

$$\text{sign}\left(\frac{\partial \log \omega_t}{\partial \log F_t}\right) = \text{sign}(\zeta - \mu) \tag{8}$$

- An increase in labor supply increases wage inequality when women compete more with low skill men than with high skill men $(\zeta > \mu)$

Implementation

- Data: Microdata from decennial CENSUS (IPUMS) - 1940-1950; 47 States.

$$\log w_{ist} = \delta_s + \gamma_{1950} + f_i + X'_{ist}\beta^g_t + \xi \log\left(\frac{F_{st}}{M_{st}}\right) + \eta f_i \log\left(\frac{F_{st}}{M_{st}}\right) + u_{ist} \quad (9)$$

- $w_{ist} \equiv$ weekly earnings.
- $X'_{ist}\beta^g_t$ gender specific observables.
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- Model disciplines interpretation of η and ξ
- $\eta = \frac{1}{\sigma_{MF}}$; $\eta + \xi = 1/\sigma_F$.
- Theoretical restriction

$$\frac{1}{\sigma_F} = -(1 - s^M_t)\alpha - (s^M_t \sigma^{-1}_{MF}) \quad (10)$$

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- Testing under "calibration": $\alpha = 0.33$; $s^M_t = 0.805$
- What happened to capital?
- What is a "market" here?
- Would you actually run this regression?

Can reject perfect substitutes and model restriction holds

TABLE 10
IMPACT OF FEMALE LABOR SUPPLY ON FEMALE/MALE EARNINGS DIFFERENTIAL, 1940–50
Dependent Variable: Log Weekly Earnings (Whites)

	Variations of the Baseline Specification			Lagged Dependent Variable	Excludes the South	Region × 1950	Ages 25–34	Hourly Earnings
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(LS_{female}/LS_{male})$	−.51	−.47	−.25	−.24	−.43	.45	−.23	−.10
	(.11)	(.09)	(.20)	(.24)	(.20)	(.29)	(.23)	(.18)
$\ln(LS_{female}/LS_{male}) \times$ female	−.31	−.25	−.42	−.35	−.61	−.27	−.20	−.50
	(.13)	(.12)	(.19)	(.15)	(.31)	(.32)	(.17)	(.22)
Implied $\sigma_{M/F}$	3.21	4.26	2.25	2.82	1.47	3.98	6.08	1.86
Estimated $\sigma_{M/F}$	3.18	3.94	2.37	2.86	1.64	3.75	5.08	2.01
Estimated σ_F	−1.21	−1.38	−1.48	−1.69	−.96	5.56	−2.34	−1.68
p-value (implied $\sigma_{M/F}$ = estimated $\sigma_{M/F}$)	.58	.51	.63	.90	.37	.84	.43	.45
	First-Stage Coefficients							
Mobilization rate × 1950	1.56	1.75	1.14	1.46	1.15	.79	1.32	1.29
	(.19)	(.22)	(.30)	(.30)	(.38)	(.32)	(.27)	(.28)
Observations			267,720		211,590	267,720	80,592	354,794
Includes share farmers, share nonwhite, and average education	no	no	yes	yes	yes	yes	yes	yes

NOTE.—Standard errors (in parentheses) account for clustering on state and year of observation. Each estimate is from a separate pooled 1940–50 micro data 2SLS regression of log weekly pooled male and female earnings on the log ratio of female to male labor supply instrumented by the WWII state mobilization rate interacted with a 1950 dummy. All models control for year main effects, veteran status (for males), marital status, years of completed education, a quartic in potential experience, and state of residence. Specifications except for col. 1 also control for female age structure and state/country of birth. All individual variables, aside from state of residence/birth, are interacted with a 1950 dummy. Variables are also interacted with a female dummy. Models are weighted by census sampling weights. See the notes to tables 2 and 9 for additional sample details.

Another estimating equation

$$\log w_{ist}^{M} = \delta_s + \gamma_{1950} + c_i + X_{ist}'\beta_t^e + \pi \log\left(\frac{F_{st}}{M_{st}}\right) + \theta c_i \log\left(\frac{F_{st}}{M_{st}}\right) + \nu \log\left(\frac{C_{st}^m}{H_{st}^m}\right) + r_{ist}$$

- Now just focus on earnings of men: college and high school graduates.
- $\sigma_{FH} = \frac{1}{\pi} \rightarrow$ cross elasticity b/w females and male high school graduates.
- $\sigma_{FC} = \frac{1}{\pi+\theta} \rightarrow$ cross elasticity b/w females and male college graduates.
- If $\frac{\pi+\theta}{\pi} = \frac{\sigma_{FH}}{\sigma_{FC}} < 1 \rightarrow$ female labor supply has greater impact on high school graduates.

Women increased inequality among men

TABLE 12
IMPACT OF FEMALE LABOR SUPPLY ON MALE EDUCATIONAL EARNINGS DIFFERENTIAL, 1940–50
Dependent Variable: Log Weekly Earnings

	Regression				
	(1)	(2)	(3)	(4)	(5)
	A. Male College/High School Graduate Log Weekly Earnings Differentials (Whites) (N=58,885)				
$\ln(LS_{female}/LS_{male})$	−.471	−.466	−.269	−.428	−.270
	(.126)	(.118)	(.228)	(.141)	(.228)
$\ln(LS_{female/male}) \times$ college	.206	.261	.233	.283	.231
	(.126)	(.117)	(.210)	(.131)	(.210)
$\ln(LS_{college\text{-}male}/LS_{HS\text{-}male}) \times$ college				−.042	−.004
				(.087)	(.080)
	First-Stage Coefficients				
Mobilization rate × 1950	1.62	1.79	1.18	1.64	1.19
	(.20)	(.23)	(.30)	(.24)	(.30)
Implied relative cross-elasticity $\sigma_{HS/CLG}$	.56	.44	.14	.34	.15
	B. Male High School Graduate/Eighth Grade Log Weekly Earnings Differential (Whites) (N=93,244)				
$\ln(LS_{female}/LS_{male})$	−.192	−.220	−.115	−.173	−.129
	(.109)	(.107)	(.205)	(.149)	(.203)
$\ln(LS_{female/male}) \times$ high school	−.279	−.246	−.154	−.153	−.140
	(.097)	(.095)	(.178)	(.117)	(.173)
$\ln(LS_{high\ school\text{-}male}/LS_{8th\text{-}male}) \times$ high school				−.070	−.145
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Ratio of cross elasticity b/w 0.4-0.6 but imprecisely estimated

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Katz and Murphy 1992

- They document massive growth in various measures of wage inequality in the late 70's / early 80's using March CPS file.
- Let's open a parenthesis here: Every labor economist must know what March CPS file actually means.
- CPS stands for the current population survey. It contains two key sources
  - March CPS: retrospective questions on annual earnings, weeks worked and usual hours per week for the year prior to the interview $\rightarrow$ typically used to study annual earnings, weeks worked and "usual" hours per week in the year prior to the interview
  - May Outgoing Rotation Group: point-in-time estimates of weekly and hourly wages in the week prior to the interview $\rightarrow$ typically used to study hourly wages
  - Some inconsistencies over the years. Lots of very smart people spent a significant amount of their time harmonizing these datasets (Autor, Katz and Kearney, 2008). That is just part of the job.

Starting point

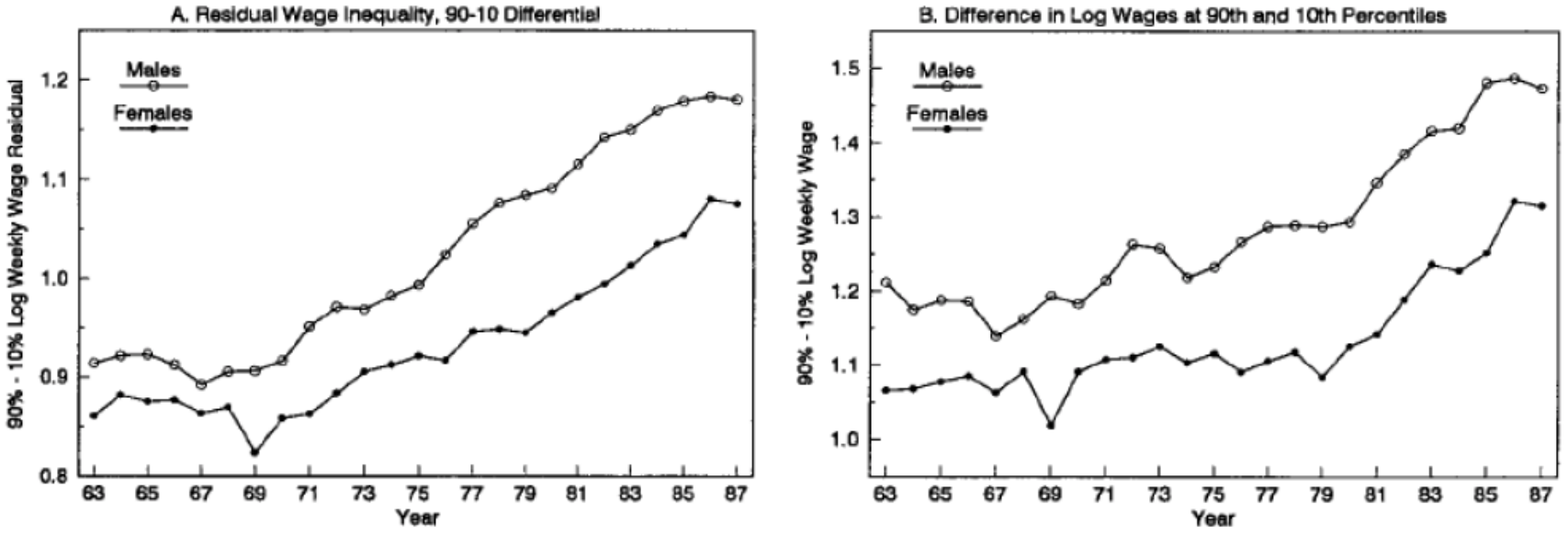
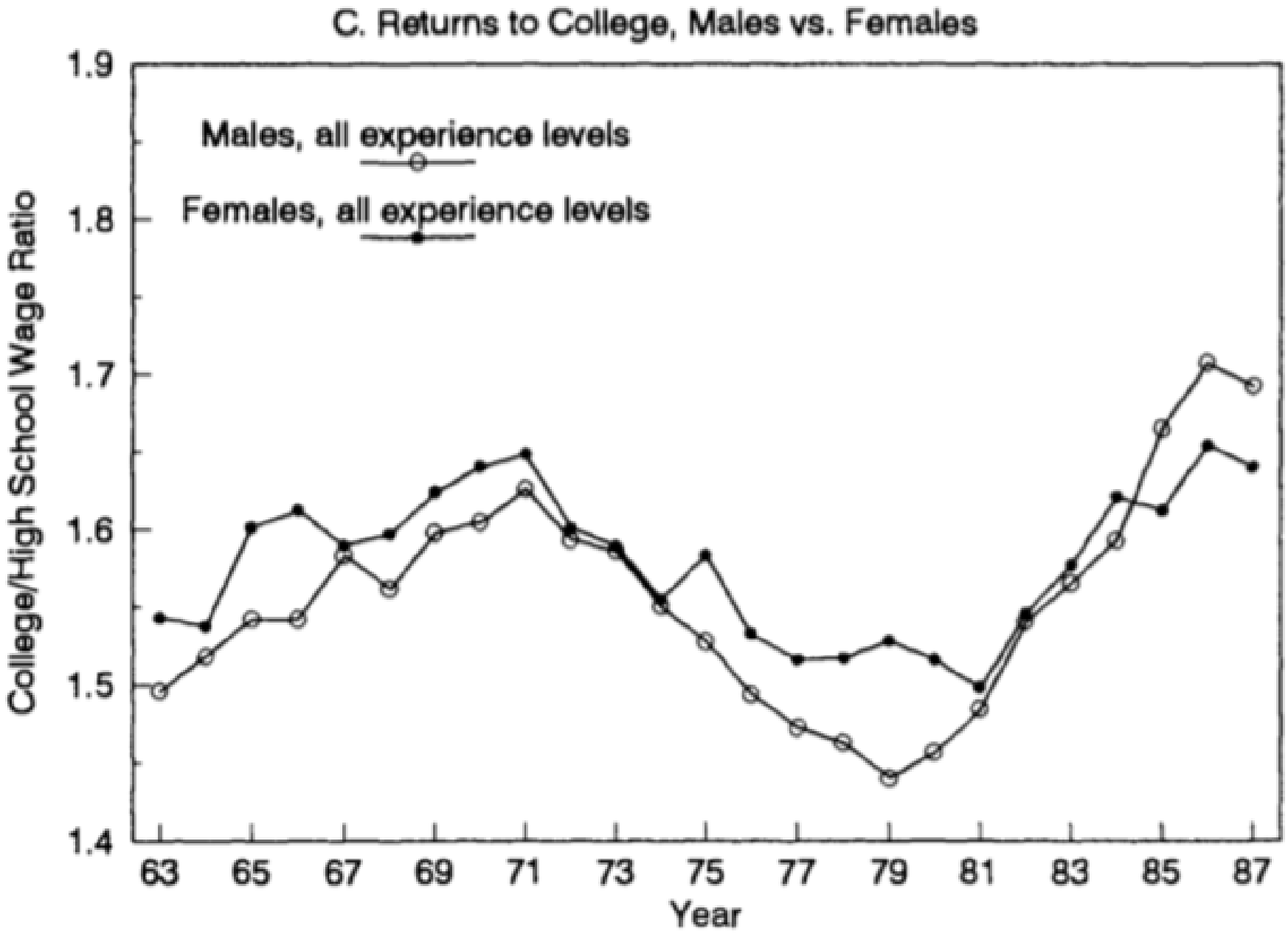


FIGURE II
Changes in Overall and Residual Wage Inequality

Residualized wage means: "residuals from a regression of log weekly earnings on a quartic in experience fully interacted with sex and four education level dummies and linear terms in education within these categories"

Starting point



Trends in Supply

TABLE II

RELATIVE SUPPLY CHANGES, 1963–1987*

Group	Change in log share of aggregate labor input (multiplied by 100) 1963–1971	1971–1979	1979–1987	1963–1987
Gender:				
Men	−2.9	−4.9	−4.2	−12.0
Women	11.2	15.7	11.2	38.2
Education (years of schooling):				
8–11	−35.2	−48.6	−41.9	−125.7
12	7.6	−4.8	−4.8	−2.0
13–15	20.3	23.3	6.7	50.3
16+	17.8	24.1	15.6	57.5

KM Idea

On the other hand, the table does illustrate the possibility that differences in the rate of growth in the relative supply of college graduates may help explain the time pattern of changes in the college wage premium. The largest increase in the supply of college graduates comes during the 1971–1979 period in which the college wage premium declined, and the smallest growth of supply comes during the 1979–1987 period in which the college wage premium expanded sharply. A smooth secular increase in the relative demand for college graduates combined with the observed fluctuations in the rate of growth of relative supply could potentially explain the movements in the college wage premium from 1963 to 1987.

Are these trends supply driven?

- Multi-factor framework. Let X_t denote all labor inputs.
- These inputs obey a labor demand law

$$X_t = D(\underbrace{W_t}_{\text{Vector of Wages}}, \underbrace{Z_t}_{\text{Vector of Demand Shifters}}) \tag{11}$$

- Total differentiation implies

$$dX_t = D_w dW_t + D_z dZ_t \tag{12}$$

- Premultiply by dW_t' we get

$$dW_t^{'}(dX_t - D_z dZ_t) = dW_t' D_w dW_t \tag{13}$$

- Can we sign the right-hand-side?

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$$dW_t^{'}(\underbrace{dX_t - D_z dZ_t}_{\text{"Net" Change in Inputs}}) = dW_t' D_w dW_t \leq 0 \tag{14}$$

- Now suppose demand shifters did not change over time ($dZ_t = 0$), then we have the implication that for any time period t and τ

$$(W_t - W_\tau)(X_t - X_\tau) \leq 0 \tag{15}$$

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- KM evaluate to what extent changes in relative wage change is explained by changes in relative supplies.

Lots of wrong signed estimates, especially in 1985

TABLE III

INNER PRODUCTS OF CHANGES IN RELATIVE WAGES WITH CHANGES IN RELATIVE QUANTITIES FOR 64 DEMOGRAPHIC GROUPS

5-year centered interval	5-year centered interval			
	1965	1970	1975	1980
Inner products of actual changes:				
1970	0.0128			
1975	−0.1129	−0.1084		
1980	−0.0893	−0.1605	−0.0040	
1985	0.3813	0.1704	0.2224	0.1421
Inner products of changes in detrended data:				
1970	−0.0251			
1975	−0.0423	−0.0351		
1980	0.0074	−0.0201	−0.0070	
1985	−0.0028	−0.0037	−0.0402	0.0138

Closing the Circle

- Conclusion: something must have happened to the relative demand for skill $\rightarrow$ Skill biased technological change.

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- Conclusion: something must have happened to the relative demand for skill $\rightarrow$ Skill biased technological change.
- Use a CES production function

$$F_t(H_t, L_t) = (A_{H_t} H_t^{\rho} + A_{L_t} L_t^{\rho})^{\frac{1}{\rho}} \tag{16}$$

- Remember from previous lecture that $\sigma = \dfrac{1}{1-\rho}$
- $(A_{H_t}, A_{L_t}) \equiv$ technology parameters that measure the productivity of high and low skilled labor.
- In perfectly competitive markets $P = MPL$, therefore

$$\omega_t \equiv \frac{w_{H_t}}{w_{L_t}} = \frac{A_{H_t}}{A_{L_t}} \left(\frac{H_t}{L_t} \right)^{\rho - 1} \tag{17}$$

- This implies that the wage gap between high and low skilled workers is given by

$$\log \omega_t = D_t - \frac{1}{\sigma} \log \frac{H_t}{L_t} \tag{18}$$

  where $D_t = \log(\frac{A_{H_t}}{A_{L_t}})$
- KM take this model to the data parametrizing D_t as a linear trends

Closing the Circle

- Conclusion: something must have happened to the relative demand for skill $\rightarrow$ Skill biased technological change.
- Use a CES production function

$$F_t(H_t, L_t) = (A_{H_t} H_t^{\rho} + A_{L_t} L_t^{\rho})^{\frac{1}{\rho}} \tag{16}$$

- Remember from previous lecture that $\sigma = \dfrac{1}{1-\rho}$
- $(A_{H_t}, A_{L_t}) \equiv$ technology parameters that measure the productivity of high and low skilled labor.
- In perfectly competitive markets $P = MPL$, therefore

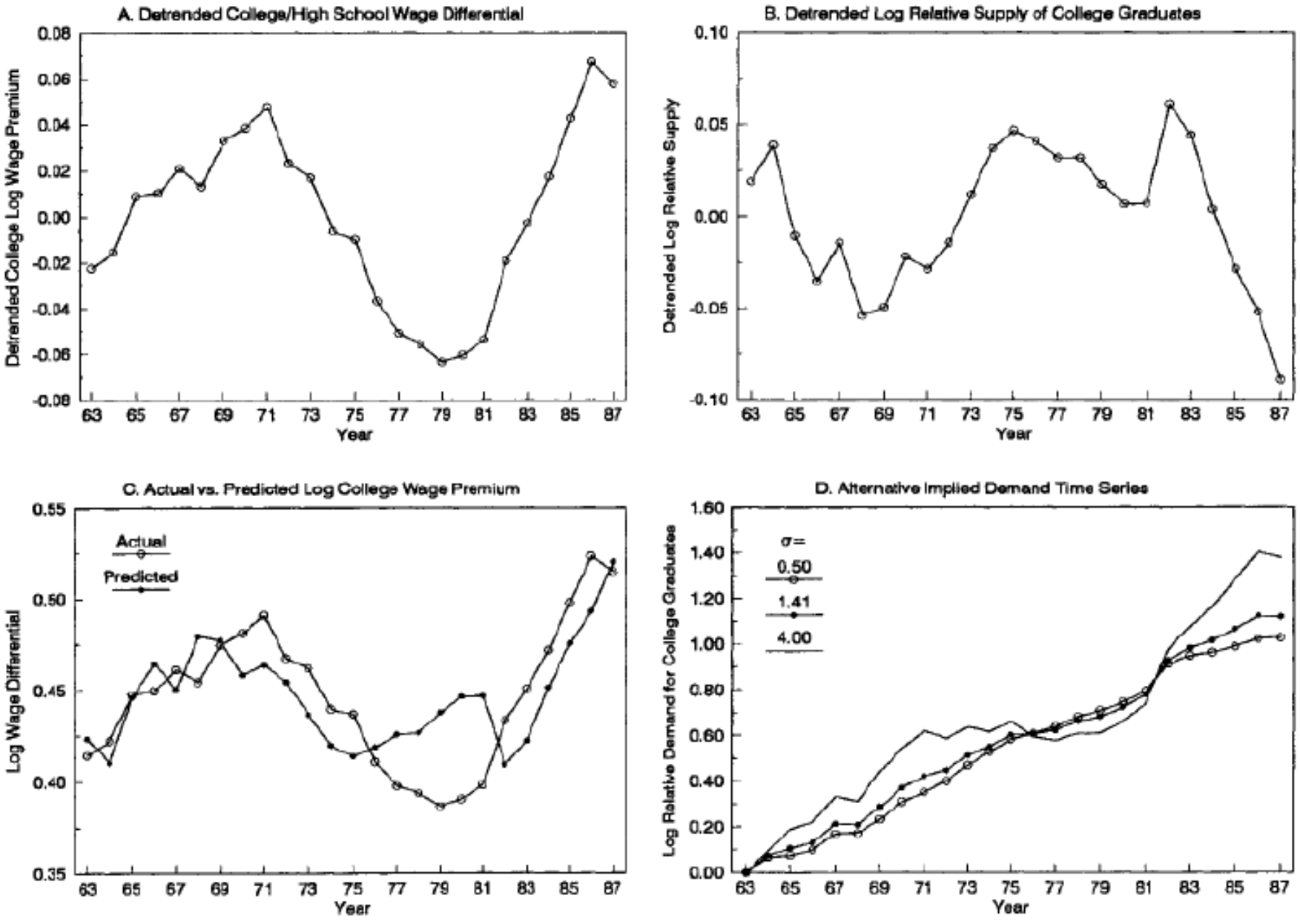
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- KM take this model to the data parametrizing D_t as a linear trends
- Wait, aren't we regressing price on quantity?(answer: yes).

$\sigma = 1.4$ is the magical number!



Wrap up

- KM is the triumph of the supply and demand paradigm.
- Incredibly influential (7.5K GS cites)
- Skill biased technological change is important to explain changes in wage inequality.

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- KM is the triumph of the supply and demand paradigm.
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- Clearly, there are several limitations
  - Demand is unobserved ("name the residual").
  - No mentioning of labor market institutions.
  - What about imperfect competition?

Bound and Johnson (1992)

- Demand is clearly important.
- Use industry / demography structure to quantitatively decompose wage gaps into components attributable to Supply / Demand / Institutions.
- Key distinction: other existing theories are not ruled out informally (like in KM) but all alternative theories are encompassed within the same approach!

Conceptual Framework

- Production in industry j is a function of I demographic groups (age $\times$ gender $\times$ education)

$$Q_j = a_j \left[\sum_{i=1} \delta_{ij} \left(b_{ij} N_{ij} \right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \tag{19}$$

where a_j is a "neutral" shifter in a given industry; b_{ij} is a skilled biased shifter; δ_{ij} : "*institutions*".

- Industry product demand obeys

$$\frac{Q_j}{Q_r} = \theta_j P_j^{-\epsilon} \tag{20}$$

- Hire until marginal revenue product equals the wage $\rightarrow$ (competitive) wage changes are given by

$$\log dW_{ic} = \left(1 - \frac{1}{\sigma}\right) d \log \underbrace{b_i}_{\text{SBTC}} - \frac{1}{\sigma} d \log \underbrace{N_i}_{\text{Supply}} + \frac{1}{\sigma} d \log \underbrace{D_i}_{\text{Demand}} \tag{21}$$

  - $N_i = \sum_j N_{ij} \rightarrow$ Total employment
  - $b_i = \frac{1}{J} \sum_j b_{ij} \rightarrow$ technical change for group i.
  - $D_i = \sum_{j=1}^{J} \delta_{ij}^{\sigma} \left(\frac{b_{ij}}{b_i} \right)^{\sigma-1} x_j$
  - $x_j = a_j \theta_j^{\frac{\sigma}{\epsilon}} Q_j^{1-\frac{\sigma}{\epsilon}}$

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Observed wages may deviate from actual wages because of institutional factors (i.e. unionizations which may vary across industries).

Identification

- Assume that SBTC and Demand follow a linear trend
- Use second differences to identify σ:

$$d^2 \log W_i = constant - \frac{1}{\sigma} d^2 \log N_i \tag{22}$$

- BJ estimate $\hat{\sigma} = 1.7$. Then use model structure to back up importance of *SBTC* and Demand factors:

$$d \log W_i = \left(1 - \frac{1}{\hat{\sigma}}\right) d \log \underbrace{b_i}_{\text{SBTC}} - \frac{1}{\hat{\sigma}} d \log \underbrace{N_i}_{\text{Supply}} + \frac{1}{\hat{\sigma}} d \log \underbrace{D_i}_{\text{Demand}} \tag{23}$$

- Industry fixed effects used to infer importance of "rents" (Krueger and Summer, 1988).

Conclusion: Supply and SBTC extremely important

TABLE 6—DECOMPOSITION OF ESTIMATED SOURCES OF 1973–1979 AND 1979–1988 RELATIVE WAGE CHANGES

Comparison groups		Relative wage change (i)	Source of relative wage change					
			Rents (ii)	Supply (iii)	Demand (iv)	Technical change: Specific (v)	Technical change: General (vi)	Unexplained (vii)
A. 1973–1979:								
College/high school	men	−0.035	0.007	−0.117	−0.008	0.009	0.073	0.001
	women	−0.073	−0.002	−0.098	−0.052	0.002	0.120	−0.043
High school/dropout ($X < 30$)	men	−0.006	0.006	−0.174	0.000	−0.007	0.153	0.016
	women	−0.002	−0.013	−0.181	0.016	−0.004	0.158	0.022
Old/young (noncollege)	men	−0.004	0.003	0.088	−0.006	0.016	−0.104	−0.001
	women	0.058	0.011	0.121	−0.007	0.024	−0.076	−0.015
Women/men		0.016	0.013	−0.066	0.003	−0.013	0.086	−0.007
B. 1979–1988:		(viii)	(ix)	(x)	(xi)	(xii)	(xiii)	(xiv)
College/high school	men	0.163	0.036	−0.100	0.013	0.019	0.196	−0.001
	women	0.118	0.015	−0.191	−0.047	0.009	0.270	0.062
High school/dropout ($X < 30$)	men	0.072	0.002	−0.193	−0.003	0.018	0.267	−0.019
	women	0.060	0.011	−0.177	0.027	0.035	0.202	−0.038
Old/young (noncollege)	men	0.107	0.015	0.062	−0.022	0.062	−0.023	0.013
	women	0.043	0.013	−0.007	−0.016	0.030	−0.006	0.029
Women/men		0.076	0.005	−0.094	0.026	−0.021	0.145	0.015

Wrap-up

- Conceptual framework to quantify competing forces in explaining wage inequality trends.
- Industry level analysis suggests demand is not that important $\rightarrow$ key is SBTC.
- Lot of important follow up to this work
  - Capital skill complementarities (Berman, Bound, Grichiles, 1994; Berman, Bound, Machin, 1998; Krusell, Ohanion, Rios-Rull, Violante, 2000)
  - Correlation with computer usage (Krueger 1993; Autor, Katz and Krueger 1993).
- End of 1990's: near consensus on importance of SBTC.

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- End of 1990's: near consensus on importance of SBTC.
- By mid 2000's: dissensus has emerged. East-West coast war begins
  - Di Nardo, Fortin and Lemieux, 1996; Card-Lemieux (2001). Card, Di Nardo (2002).
  - Institutions more important than originally accounted for.